AL DHAFRA HEALTH FORUM · ACCURATE DIAGNOSIS — التشخيص الدقيق

Diagnostic Intelligence *Lab*

“Before we trust a diagnosis, we must understand how it was reached.”

May Ahmed Mourad · Grade 12
Shamsa Faris Almazrouei · Grade 12
Fatima Mohamed Almazrouei · Grade 11

ADNOC Schools — Madinat Zayed

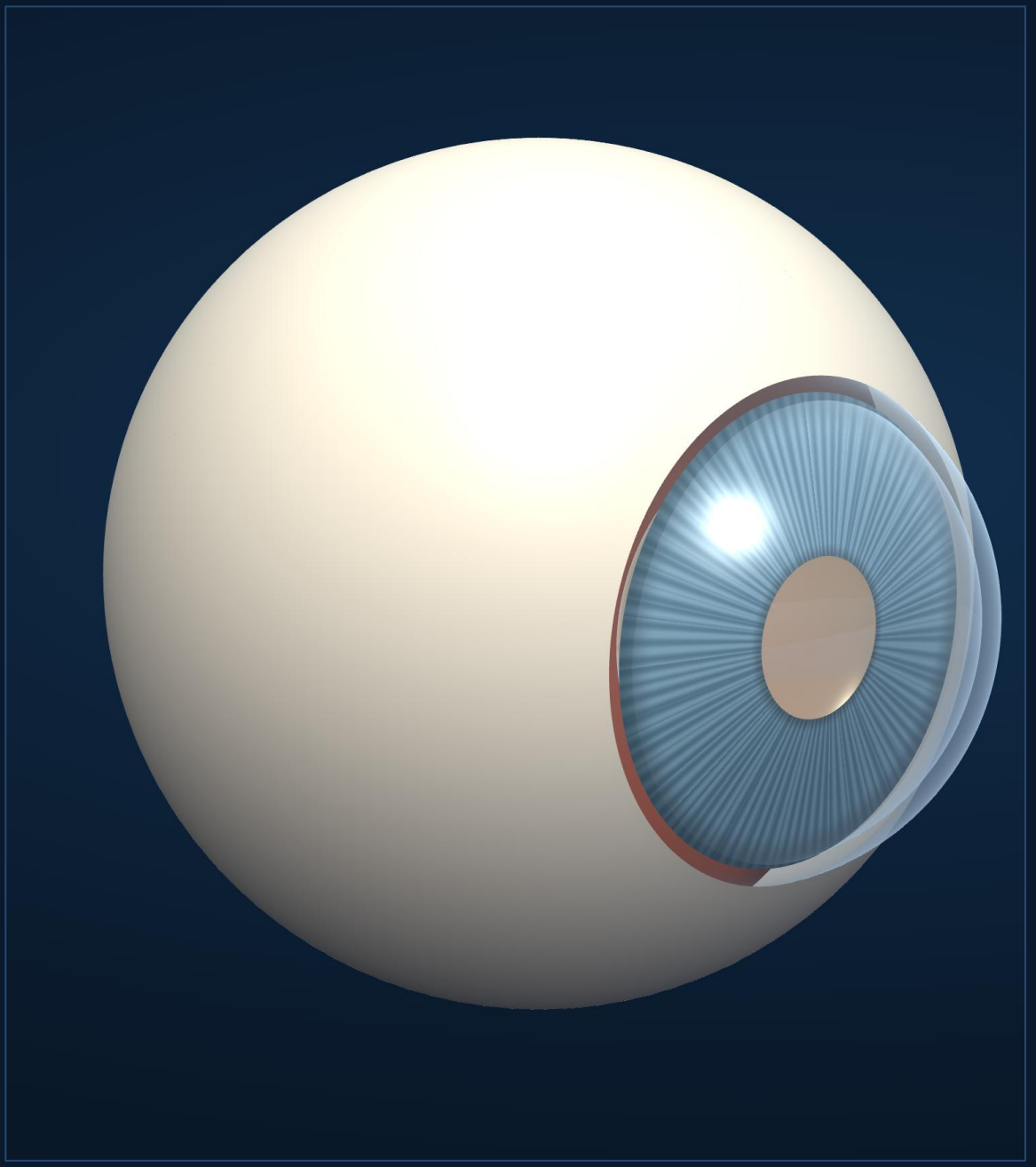
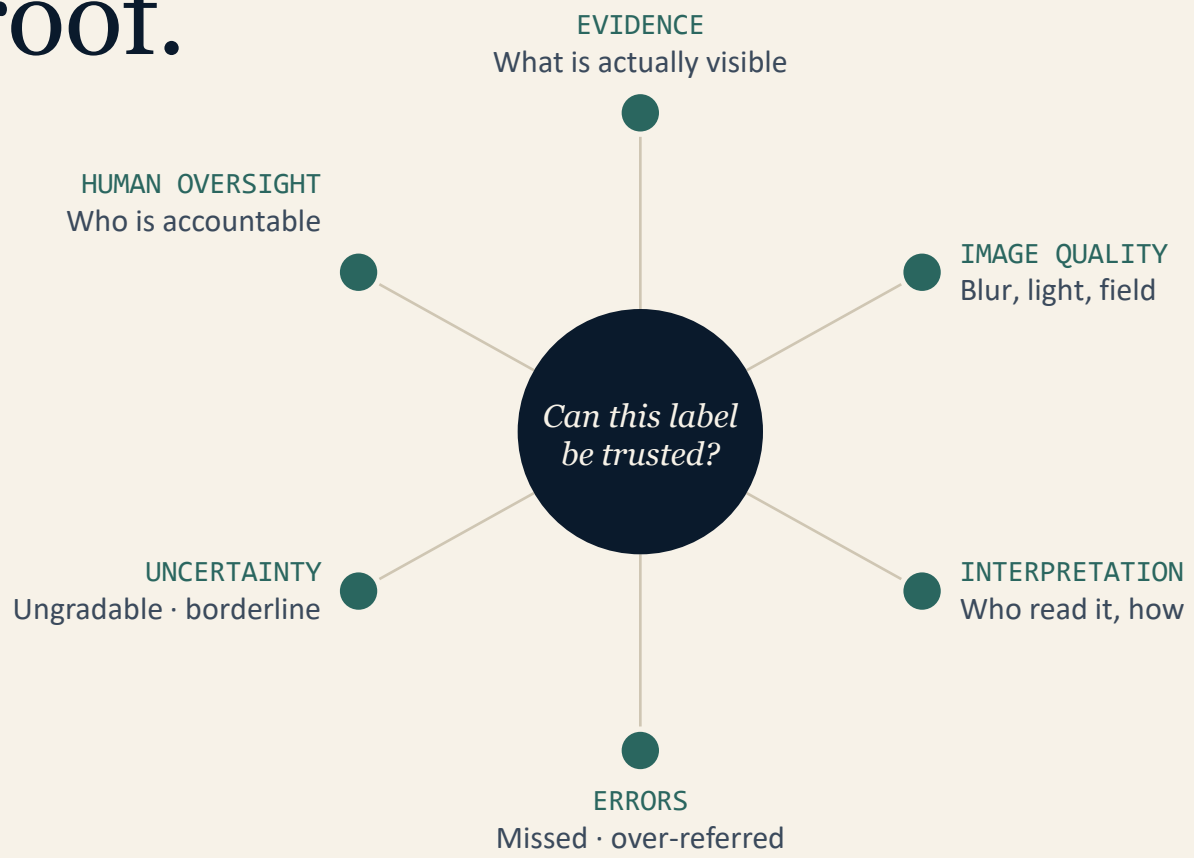


FIG. 01 · original WebGL eye · schematic, not to scale

A prediction is not proof.

The challenge is not producing a diagnostic label.

It is establishing whether that label can be trusted.



22.3%

of adults with diabetes worldwide have diabetic retinopathy (95% CI 19.7–25.0)

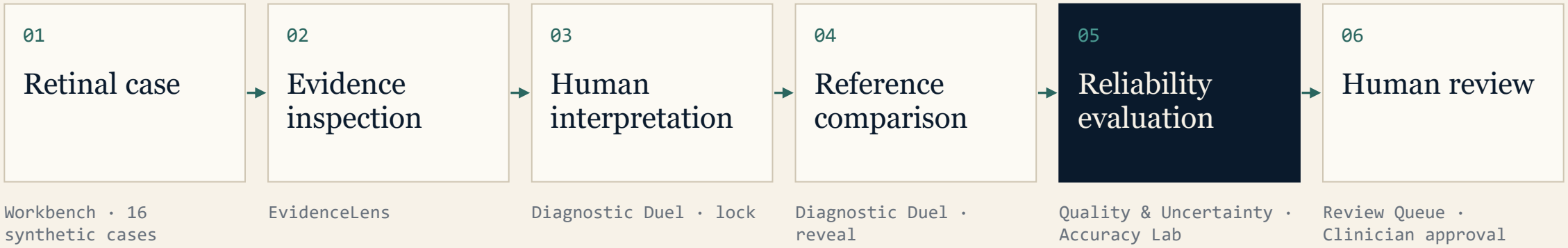
Teo et al., Ophthalmology 2021 · 59 studies

≈ 1.3 M

adults (20–79) with diabetes in the UAE in 2024

IDF Diabetes Atlas · UAE country report

How accurately is a diagnosis read — and how can we measure that?



THE PLATFORM INVESTIGATES

Diagnostic interpretation

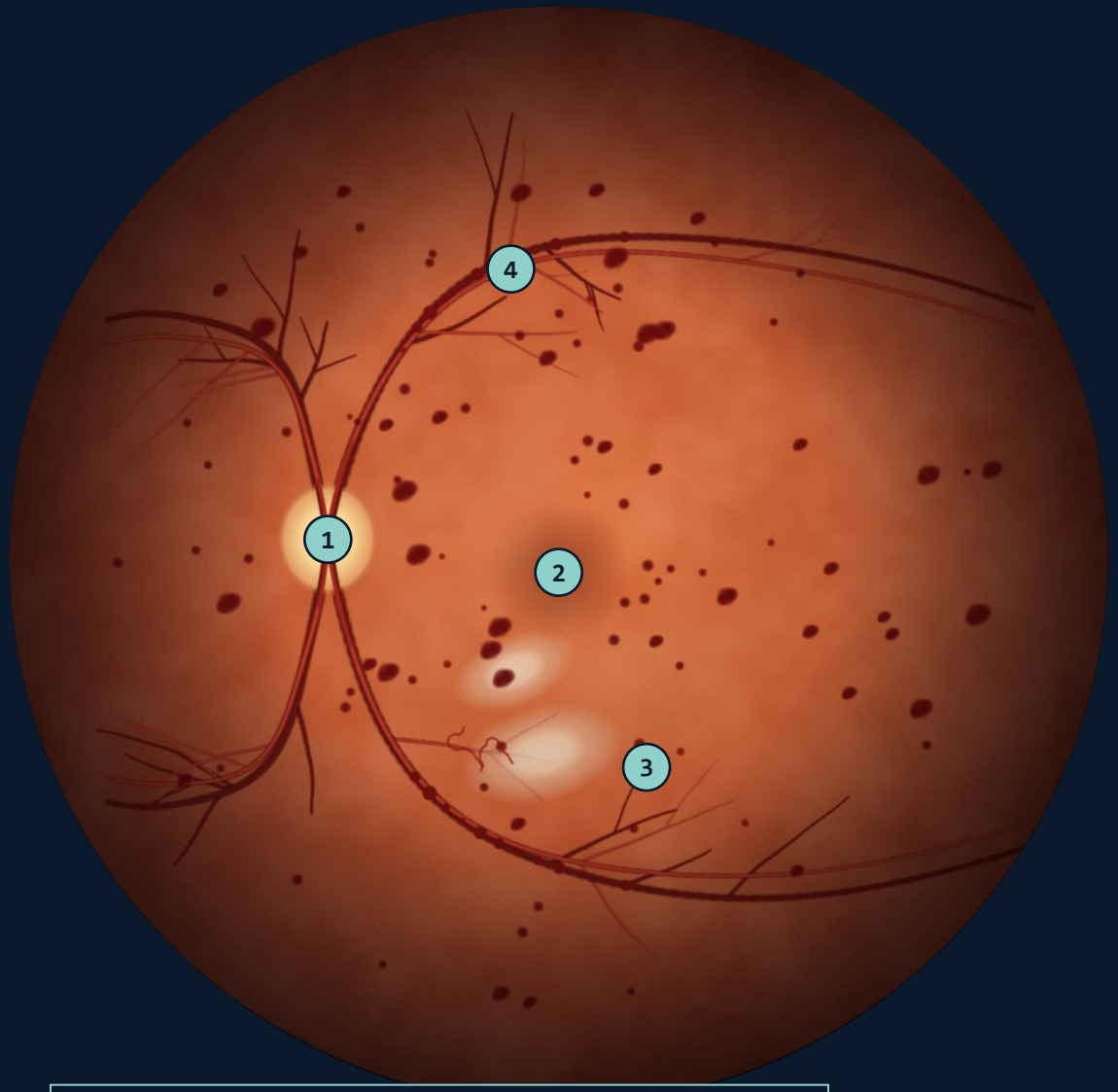
Evidence inspection

Error detection

Reliability

Accuracy evaluation

Accountability



SYNTHETIC ILLUSTRATION · NOT A PATIENT

Retinal imaging gives diagnosis a structured case study.

- **Visible evidence.** Lesions can be seen, counted, located.
- **A clear question.** Refer — or re-screen.
- **Two unequal errors.** False negative · false positive.

1 OPTIC DISC · nasal to the macula
2 MACULA · FOVEA · sharpest vision
3 HAEMORRHAGE · authored reference annotation
4 VENOUS BEADING · a severe-grade sign

ICDR SEVERITY · AUTHORED REFERENCE LABELS

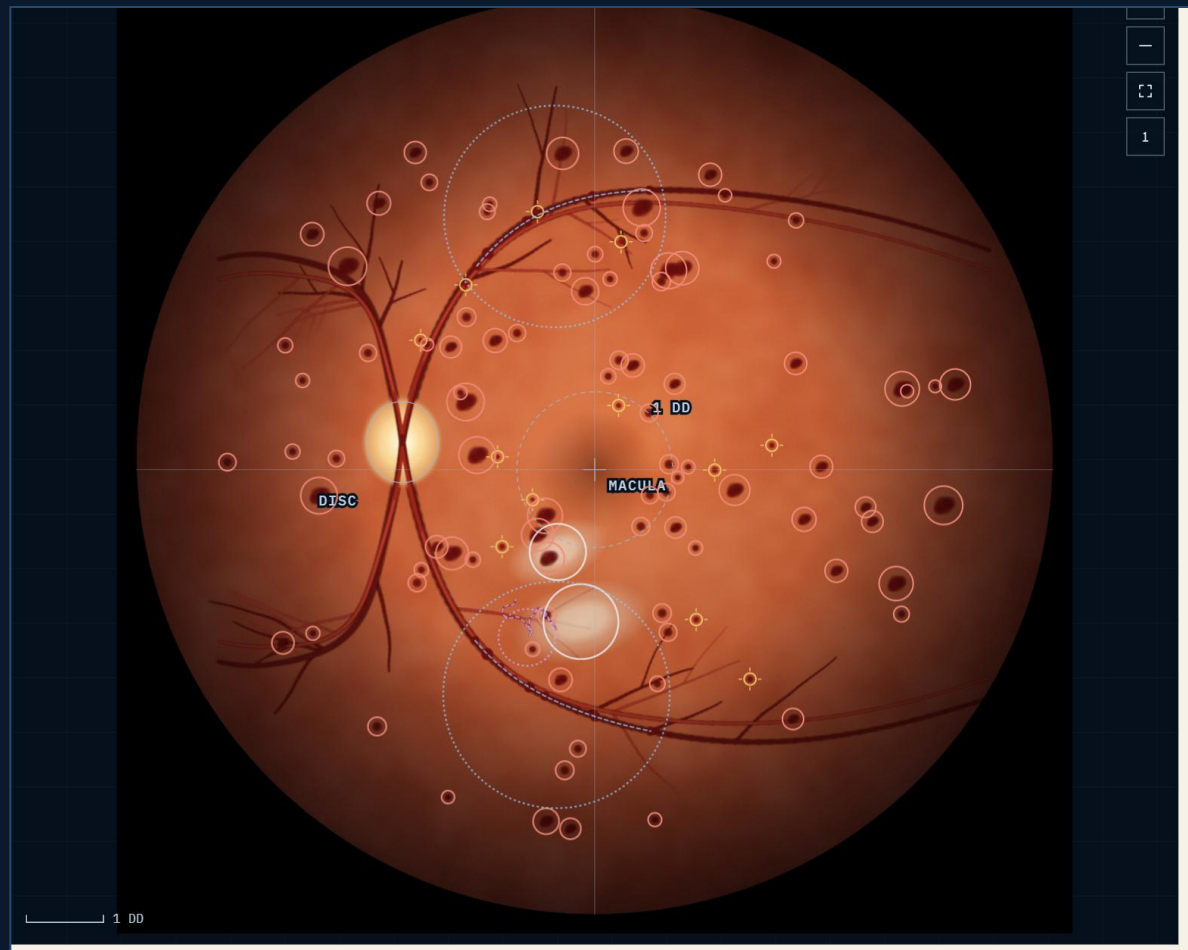
0	1	2	3	4

NON-REFERABLE — REFERABLE · or suspected DME

Retinal screening is the first research use case — not the limit of the platform.

ICDR: Wilkinson et al., Ophthalmology 2003 · ETDRS Report 10, 1991

Evidence first. Answer later.



Anatomy guides
Optic disc outline, macular centre, 1 disc-diameter (DD) macular zone, quadrant axes. Orientation only.

REFERENCE ANNOTATION LAYERS

- Microaneurysm 12
- Intraretinal haemorrhage 88
- Cotton-wool spot 2
- Venous beading 2
- IRMA 1

Show Hide

OVERLAY OPACITY

DRAGGABLE COMPARISON

Off | Plain | annotated | Colour | green-channel enhanced

Green-channel contrast stretch: a standard preprocessing view. It changes display only; nothing is annotated by it.

FIG. 10 SYNTHETIC TEACHING ILLUSTRATION · C10 · LEFT EYE (OS)

Scroll or pinch to zoom · drag to pan · +/− keys · 0 to fit · click an annotation for details

— ZOOM · PAN · NATIVE RES.
Wheel, pinch or keys; zoom level shown

— ANATOMY GUIDES
Disc outline · 1-DD macular zone · quadrants

— LAYERS + OPACITY
Annotation layers revealed one at a time

— GREEN-CHANNEL COMPARISON
Draggable plain | enhanced divider

BLIND-FIRST SAFEGUARD

Reference annotations stay locked until the interpretation is locked.

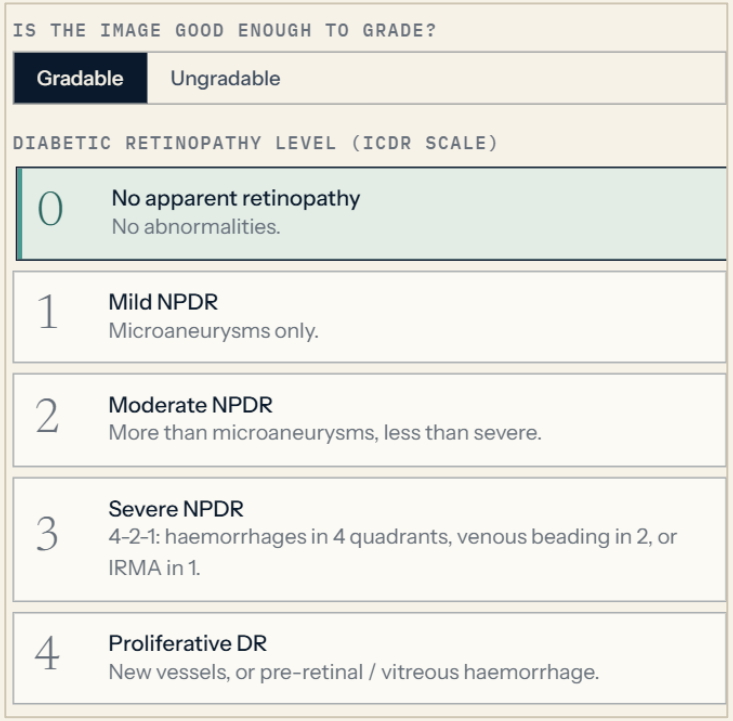
Reduces anchoring in the educational reading. Early access = Study Mode = practice only.

LIVE APPLICATION · EvidenceLens · synthetic case C10 opened in Study Mode

Read. Lock. Reveal.

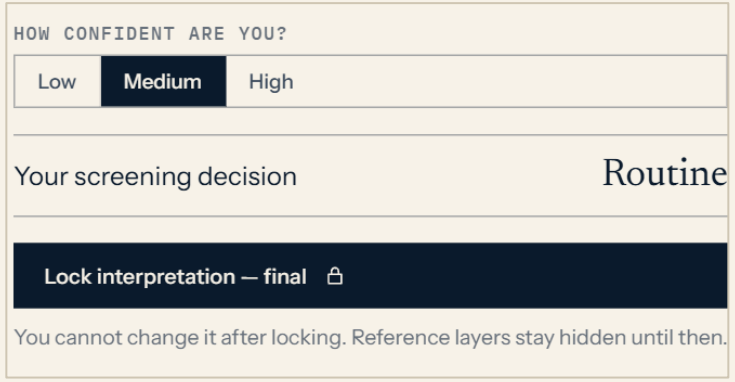
EDUCATIONAL HUMAN-VERSUS-REFERENCE COMPARISON. No trained diagnostic AI model is involved.

READ



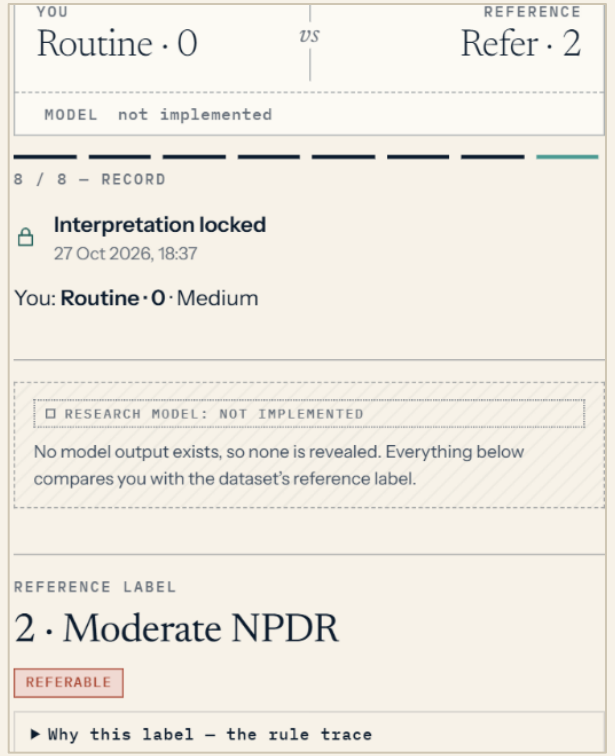
1 Observe · 2 Gradability · 3 Confidence · 4 Interpret

LOCK



5 Lock — final, no edits

REVEAL



6 Reference · 7 Discrepancies

The interpretation comes first. The reference comes afterward.

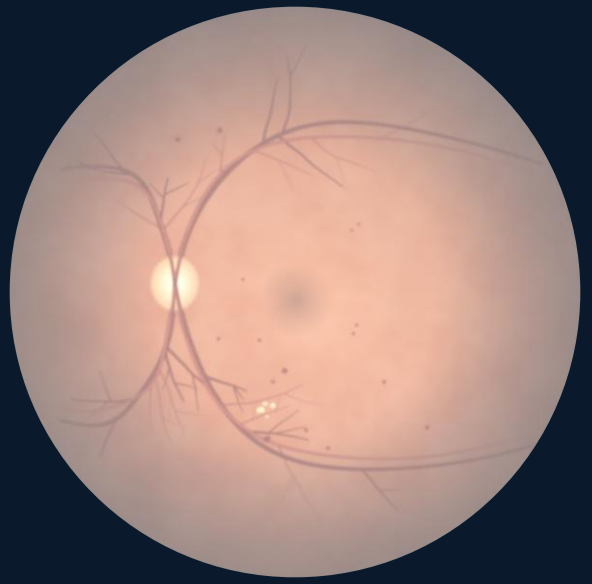
Judge the evidence before the interpretation.



ORIGINAL



BLUR σ = 7 px



WASHED-OUT PRESET

	ORIGINAL	BLUR σ 7	WASHED-OUT
Sharpness	3.48	0.98	1.91
Brightness	76.58	76.54	171.12
Contrast	32.58	31.83	17.87
Verdict	ADEQUATE	INSUFFICIENT	BORDERLINE

Values: the prototype’s own quality functions on case C06 (768 px analysis render), recorded by capture.mjs

MEASURED FROM PIXELS

Sharpness · brightness · contrast · evenness · field coverage

NO INVENTED CONFIDENCE

No model, so no model response to degradation is shown.

NO-RESULT PATHWAY

Insufficient quality → no grade → recapture.

HEURISTICS

Tuned on 16 teaching images; not clinically validated.

Measured, not claimed.

REFERENCE \ PREDICTED	Referable	Non-referable
Referable	**TP** TRUE POSITIVE	**FN** FALSE NEGATIVE · MISSED
Non-ref.	**FP** FALSE POSITIVE · OVER-REFERRAL	**TN** TRUE NEGATIVE

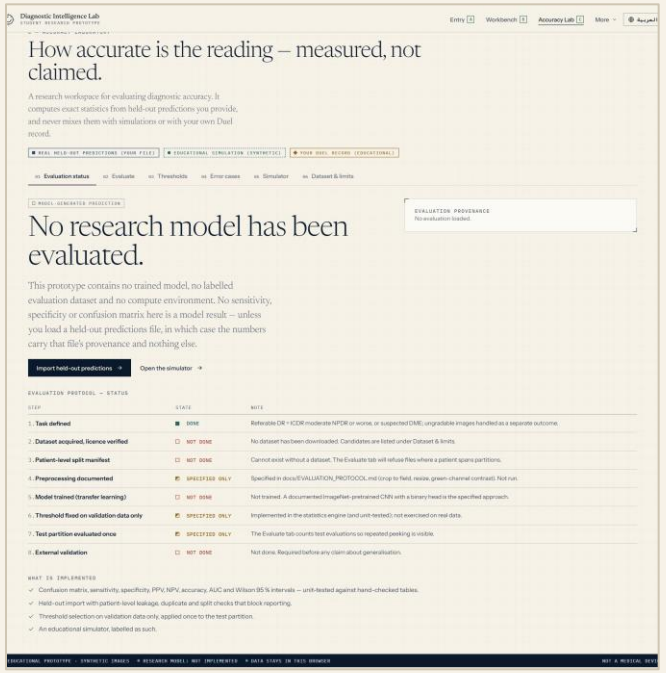
DIAGRAM — NOT A RESULT. NO MODEL HAS BEEN EVALUATED.

THE LABORATORY COMPUTES

Sensitivity	Specificity
PPV · NPV	Accuracy
ROC · AUC	Wilson 95% CI
Threshold analysis	Error cases

AND IT PROTECTS THE EVALUATION

- Patient-level leakage refused
- Threshold chosen on validation only
- Test evaluated once · counted
- Provenance: file SHA-256 + declared metadata



LIVE APPLICATION · actual state: no research model evaluated

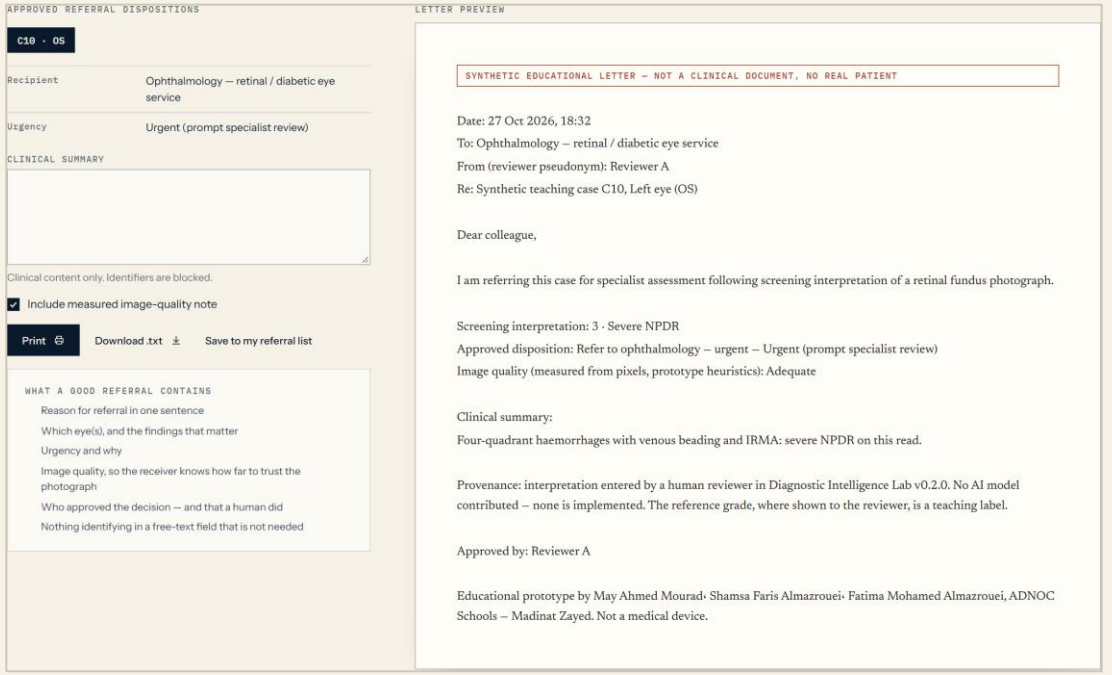
> *"The platform does not ask users to believe an accuracy claim. It establishes how that claim should be tested."*

The decision stays human.

01 Interpretation → 02 Review → 03 Approval → 04 Conditional referral draft



LIVE APPLICATION · Review Queue · discrepancy flags (sample records)



LIVE APPLICATION · CareBridge · drafted only after approval

It drafts a referral only after an eligible disposition is approved. **It does not decide urgency or treatment.**

Safeguards that are already built.

SAFETY

01

- Synthetic cases only
- No real patient uploads
- No fabricated AI predictions
- Image-quality awareness

TRANSPARENCY

02

- Blind-first interpretation
- Defined, attributed reference labels
- Evaluation provenance recorded
- Limitations stated on the page
- English and Arabic interface

ACCOUNTABILITY

03

- Human approval of every disposition
- Records stay in local browser storage
- Leakage checks protect evaluation
- Review workflows
- References, interpretations and decisions kept separate

NOT CLINICALLY APPROVED · NOT A MEDICAL DEVICE · NOT FOR REAL PATIENT CARE

The next step is not deployment.

It is rigorous validation.

TODAY	PHASE 1	PHASE 2	PHASE 3	PHASE 4	PHASE 5
Prototype • 16 synthetic cases • Evidence inspection • Interpretation workflow • Evaluation engine • Human review	**Representative data** • Licensed retinal dataset • Patient-separated train / validation / test	**Research model** • Train and evaluate • Document architecture, training, limits	**External validation** • External dataset • Subgroups · calibration • Quality robustness · errors	**Clinician-supervised research** • Performance · workflow • Automation bias • Human–AI collaboration	**Prospective evaluation** • Prospective studies • Governance · security • Regulatory requirements

Solid = implemented in the published prototype. Dashed = future research, not built.

“The future of diagnostic AI should not be judged by how confidently it predicts — but by how reliably its accuracy can be demonstrated.”

SCAN · LIVE PROTOTYPE

maymourad.github.io/**Diagnostic-Intelligence-Lab**

Open source · MIT · github.com/MayMourad/Diagnostic-Intelligence-Lab

TRANSPARENT EVIDENCE

MEASURABLE ACCURACY

RESPONSIBLE DEVELOPMENT

HUMAN OVERSIGHT

Diagnostic Intelligence Lab

May Ahmed Mourad (G12) · Shamsa Faris Almazrouei (G12) · Fatima Mohamed Almazrouei (G11)
ADNOC Schools — Madinat Zayed · Al Dhafra Health Forum · Accurate Diagnosis — التشخيص الدقيق

Four counts. Everything else follows.

Sensitivity

`TP ÷ (TP + FN)`

Of the truly referable, the share found. Misses are FN.

Specificity

`TN ÷ (TN + FP)`

Of the truly non-referable, the share correctly cleared.

PPV

`TP ÷ (TP + FP)`

When it says “refer”, how often it is right — depends on prevalence.

NPV

`TN ÷ (TN + FN)`

When it says “routine”, how often it is right — depends on prevalence.

Accuracy

`(TP + TN) ÷ all`

Overall agreement; can hide a poor sensitivity when disease is rare.

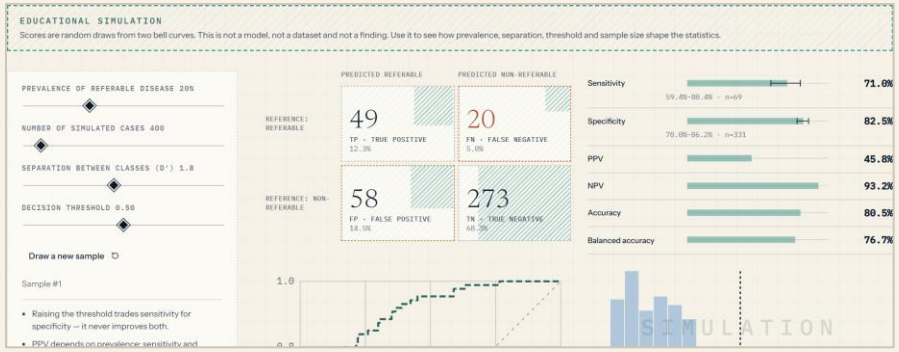
Wilson 95% CI

$\hat{p} \pm z \cdot \sqrt{(p(1-p)/n + z^2/4n^2)} \div (1+z^2/n)$

Interval for a proportion; wide when n is small.

THRESHOLD TRADE-OFF

Raising the decision threshold trades sensitivity for specificity — it never improves both. The simulator at right is stamped SIMULATION and uses random scores, not a model.



A good result on bad data is a bad result.

Held-out file
`id · patient · split · reference · score`

→

Integrity checks
`leakage · duplicates · class balance`

→

Threshold
`validation data only`

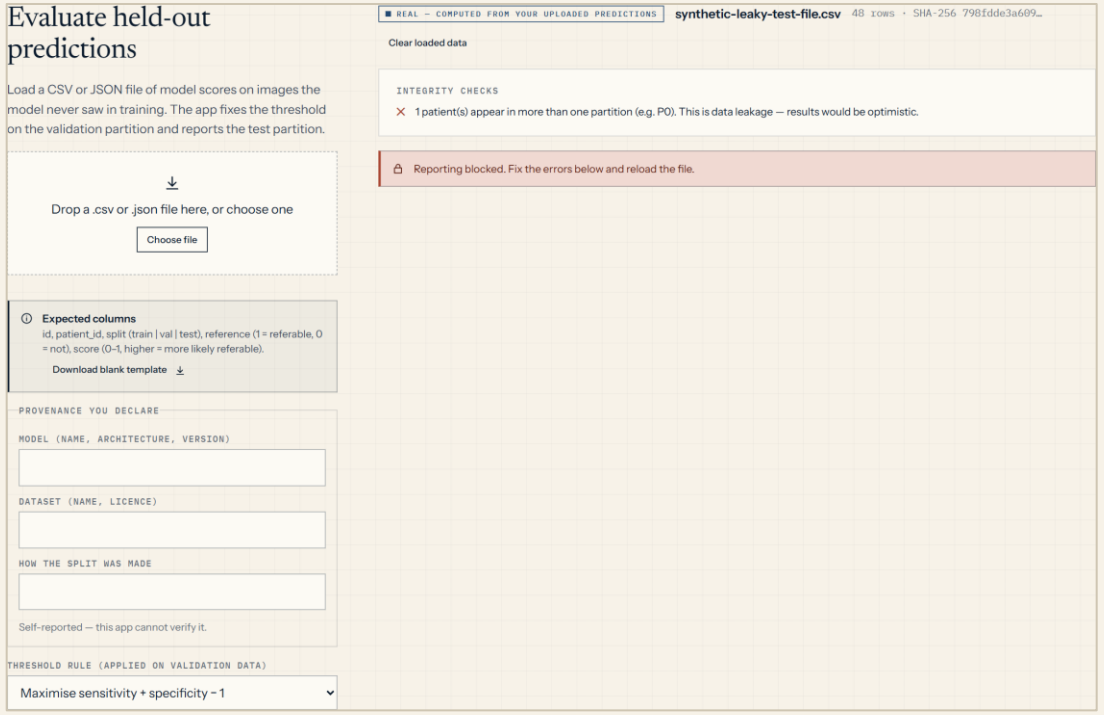
→

Test, once
`evaluations are counted`

→

Provenance
`SHA-256 · declared metadata`

- **Patient-level leakage:** one patient, one partition.
- **Separation:** train fits, validation selects, test reports once.
- **Provenance:** who made the model, on what data — declared, hashed.
- **External validation:** a different site or camera.
- **Reproducibility:** fixed seeds, unit-tested engine, CI.



LIVE APPLICATION · synthetic test file with a deliberate leak → reporting blocked

Exactly what exists — and what does not.

IMPLEMENTED

- Entry, Workbench, EvidenceLens
- Diagnostic Duel · quality and stress test
- Clinician approval · review · CareBridge
- English and Arabic interface

SYNTHETIC · EDUCATIONAL

- 16 retinal teaching illustrations
- Authored lesions → ICDR-derived labels
- Statistics simulator (stamped)
- Educational duel score

EVALUATION INFRASTRUCTURE

- Confusion matrix, sens/spec, PPV/NPV, AUC, Wilson CI
- Held-out import · leakage checks
- Validation-only thresholds
- 31 unit + 35 browser tests · CI

NOT IMPLEMENTED

- Any trained diagnostic model
- Real-dataset evaluation
- Validated sensitivity / specificity
- Evidence of improved clinician accuracy

REQUIRED BEFORE ANY CLINICAL USE

- Licensed data and ethics approval
- External and prospective validation
- Clinician-supervised research
- Governance, security, regulatory review

WHY IT IS NOT A MEDICAL DEVICE

- No validated performance claim
- Synthetic cases only; no patient data
- Never produces a diagnosis or treatment advice
- Every decision is a human’s